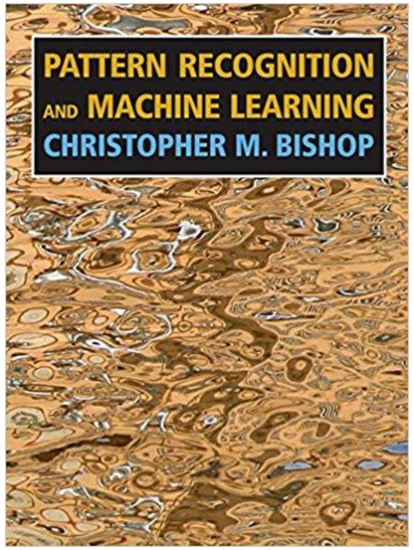
机器学习与数据挖掘

课程介绍

教材

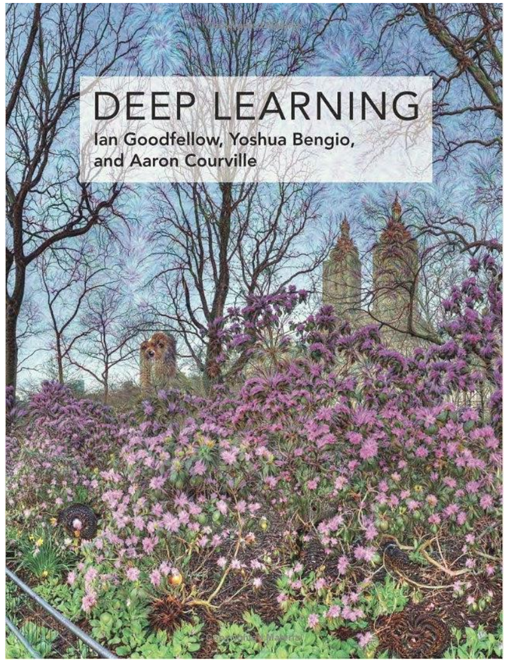
- *Pattern Recognition and Machine Learning*. Bishop. Springer, 2006



- *多模态大模型——新一代人工智能技术范式*，刘阳 林倞，电子工业出版社，2024；入选中山大学本科教材

参考书

- *机器学习*，周志华，清华大学出版社，2016
- *Deep Learning*. Goodfellow, Bengio & Courville. MIT Press, 2016



参考在线材料

- 斯坦福机器学习课程
- http://www.ml-class.org/
- http://cs229.stanford.edu/materials.html
- 斯坦福数据挖掘课程
- http://infolab.stanford.edu/~ullman/mmds.html

成绩评定标准

- 期末考试
  - ✓ 40 分
- 四个项目
  - ✓ 共 50 分: 12 + 12 + 12 + 14（期末项目）
- 课堂表现
  - ✓ 10 分

助教信息

姓名: 赵清

电话: 15322071821

E-mail: zhaoq78@mail2.sysu.edu.cn

地址: 南实验楼D302

本课程内容概览

- **机器学习基础**
  - ✓ 线性回归
  - ✓ 线性分类
  - ✓ 支持向量机
  - ✓ 决策树
  - ✓ 聚类：K-均值
  - ✓ 神经网络
- **感知&理解**
  - ✓ 线性降维：主成分分析（PCA）
  - ✓ 图像表征学习
  - ✓ 文本表征学习
  - ✓ 多模态感知与理解
- **机器生成**
  - ✓ 潜变量模型
  - ✓ EM算法
  - ✓ 变分自编码器
  - ✓ 扩散模型
  - ✓ 可控生成与AIGC
  - ✓ ………
- **数据挖掘**
  - ✓ 异常检测
  - ✓ 推荐系统
  - ✓ ……

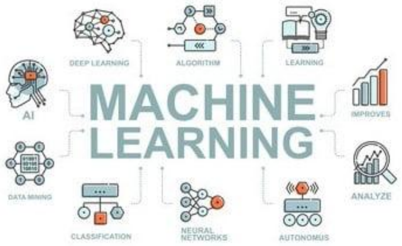
什么是机器学习？

- 学习是一个系统利用过往的经验来提升其在特定任务上表现的过程

\- Herbert Simon

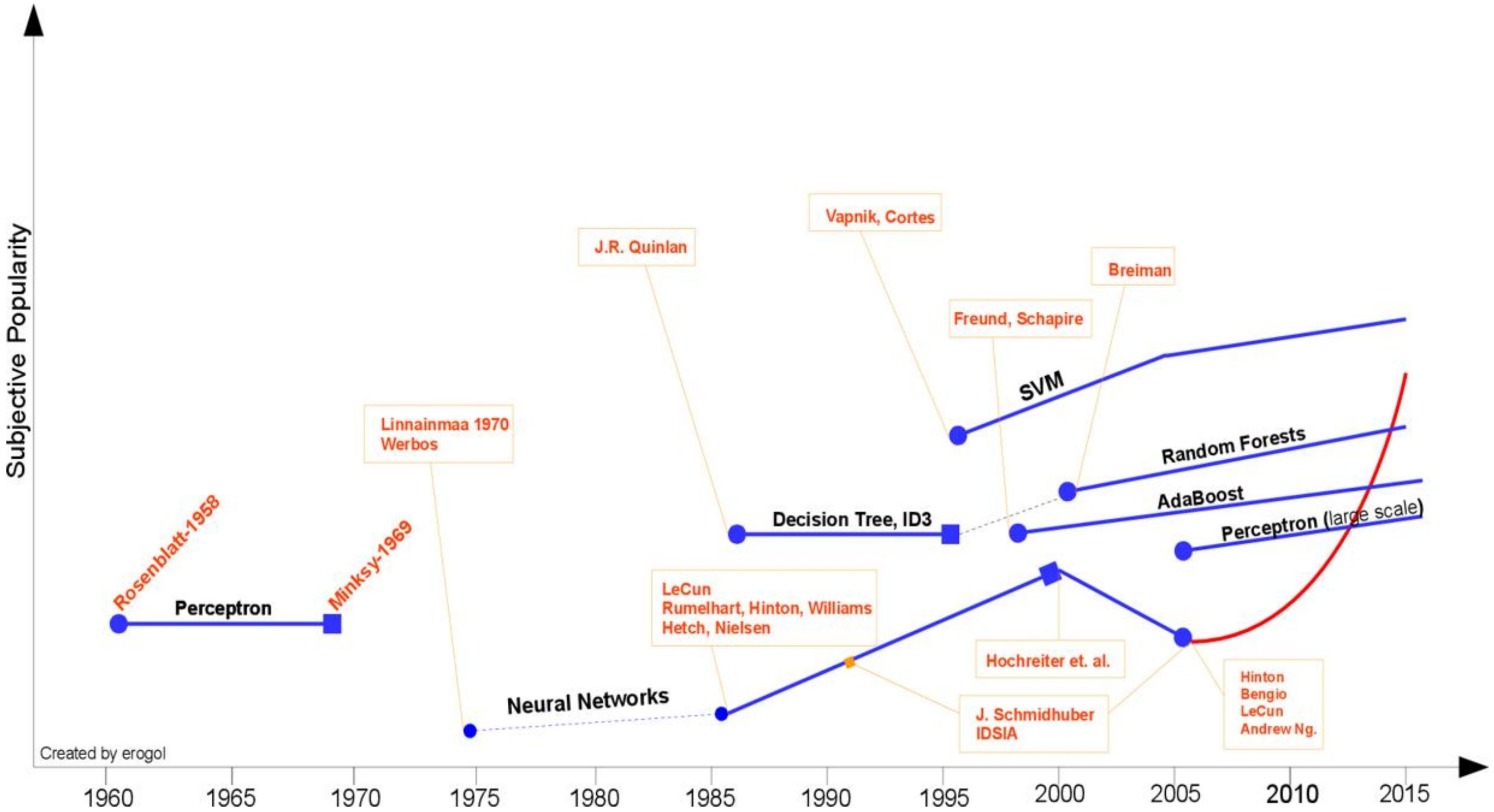
经验 → 提升

- 机器学习同样是关于如何利用过往经验来提升特定任务的表现，但其实现方式是通过*学习算法*



经验
（数据）
→ 提升

机器学习发展



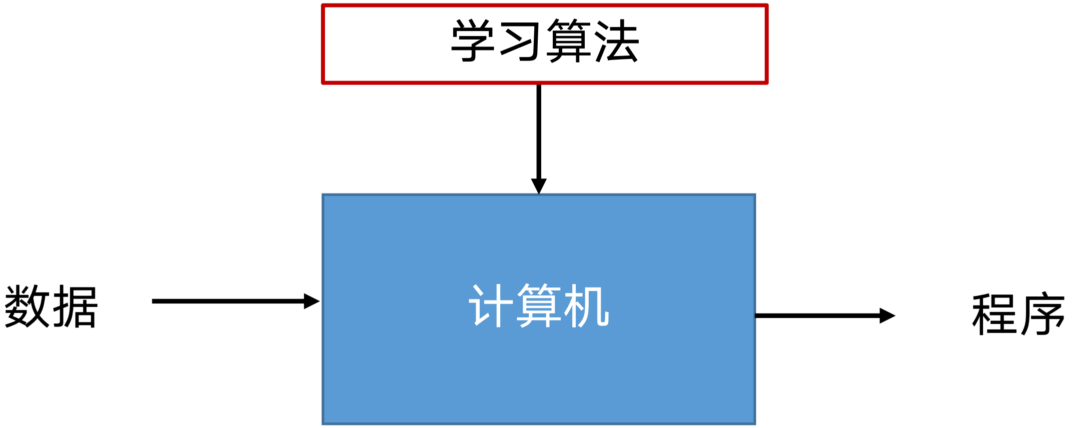
Subjective Popularity
Rosenblatt-1958
Perceptron
Minksy-1969
Linnainmaa 1970
Werbos
Neural Networks
J.R. Quinlan
Decision Tree, ID3
LeCun
Rumelhart, Hinton, Williams
Hetch, Nielsen
Vapnik, Cortes
SVM
Freund, Schapire
Breiman
Random Forests
AdaBoost
Perceptron (large scale)
Hochreiter et. al.
J. Schmidhuber
IDSIA
Hinton
Bengio
LeCun
Andrew Ng.
Created by erogol
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015

传统算法与学习算法的比较

- 学习算法与我们所熟悉的*传统算法* 有所不同

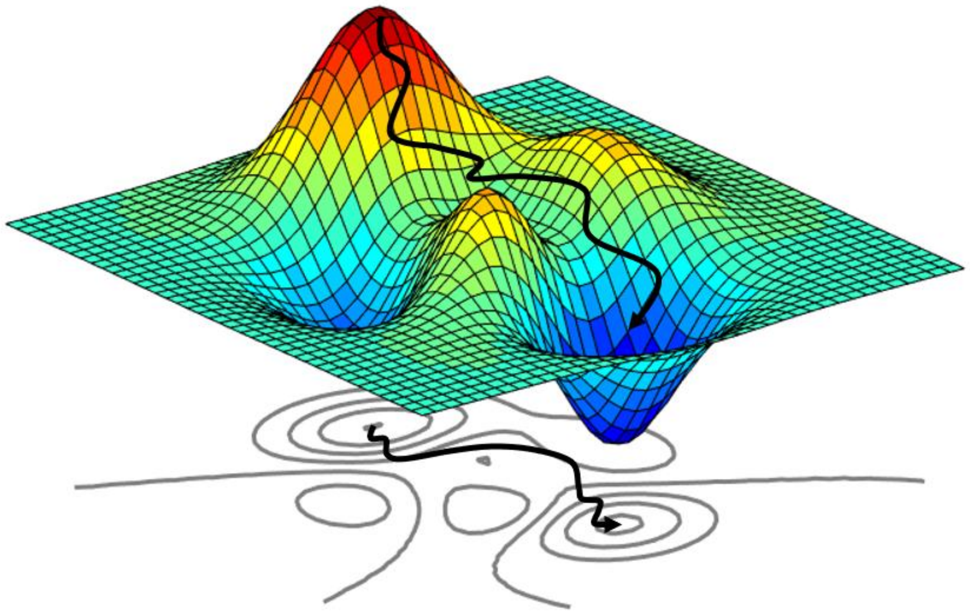
- 传统算法

- 学习算法

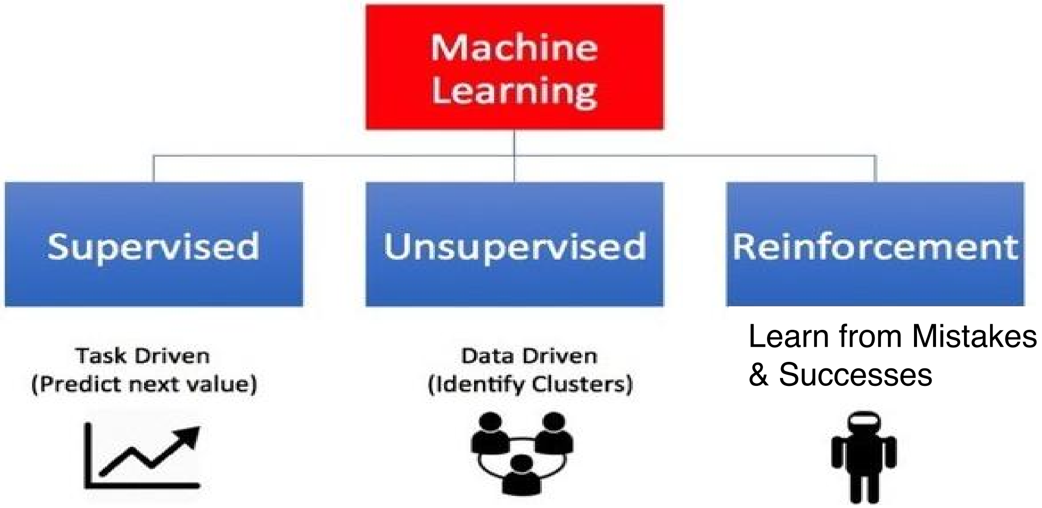


机器学习的本质

- 机器学习的核心可以看作一个优化问题
- 模型：一个带有未知参数的函数
- 策略：定义一个损失函数来衡量模型预测的好坏
- 算法：一个求解最优化问题的算法，旨在找到最优参数，使得损失函数最小化



机器学习的类型



- 监督学习
  - 给定：输入数据 + 标签（预测值）
- 无监督学习
  - 给定：输入数据

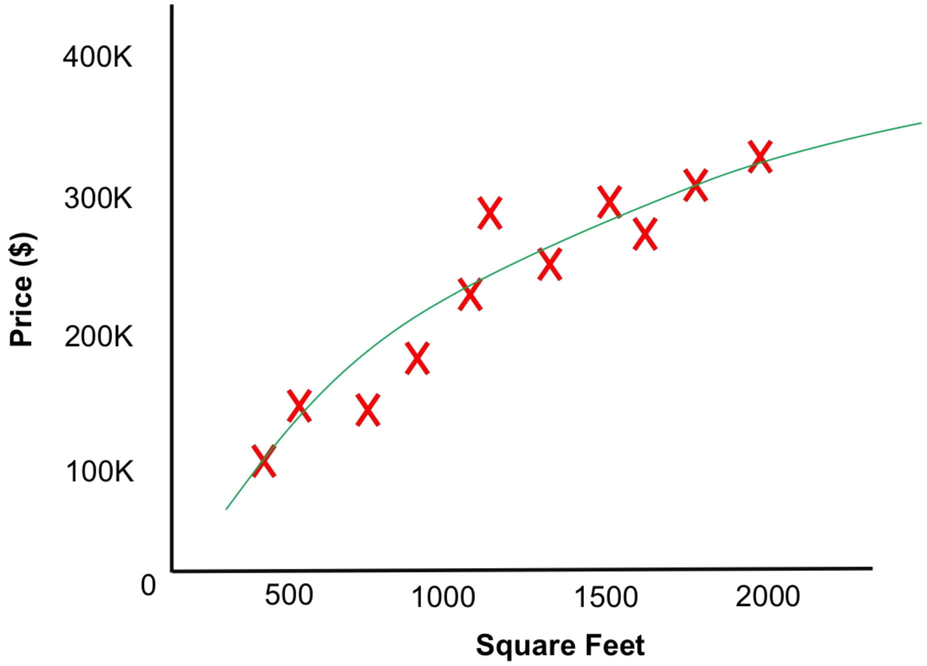
→
  - 半监督学习
  - 弱监督学习
  - 自监督学习

- 强化学习
  - 给定：一系列动作 + 奖励

监督学习

- 给定成对的输入数据和监督信号 (x_i, y_i),学习一个函数 $f(x)$以便对新的输入 x 预测其对应的 y 值

- 示例：回归

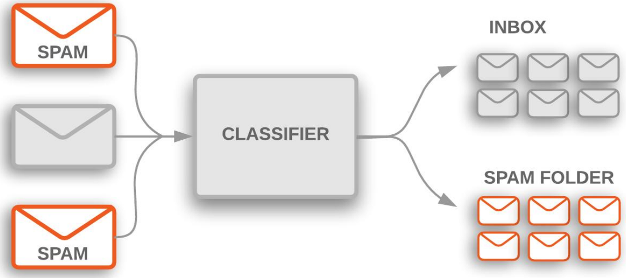


示例

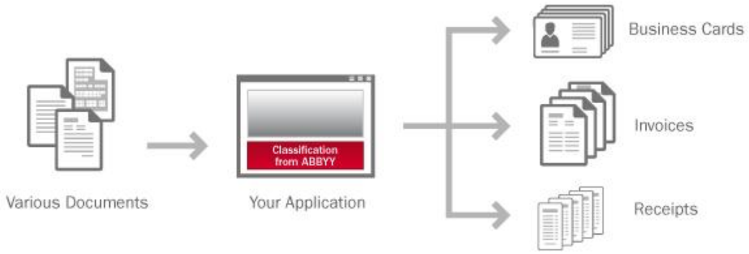
- 分类

  - 图像识别

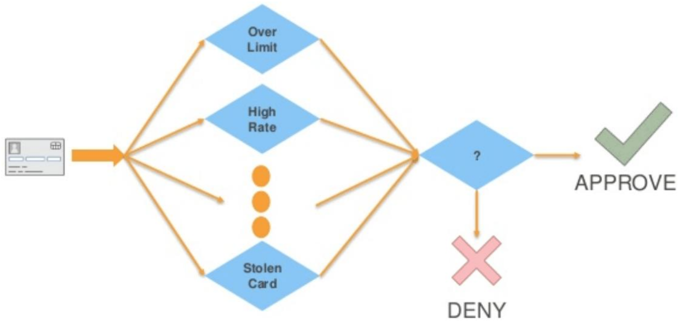
  - 垃圾邮件检测



  - 文档分类

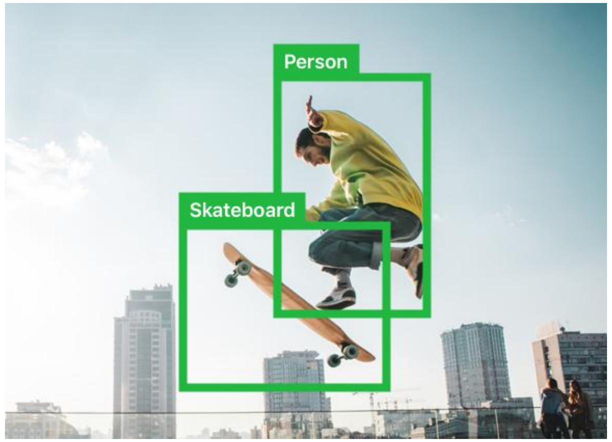


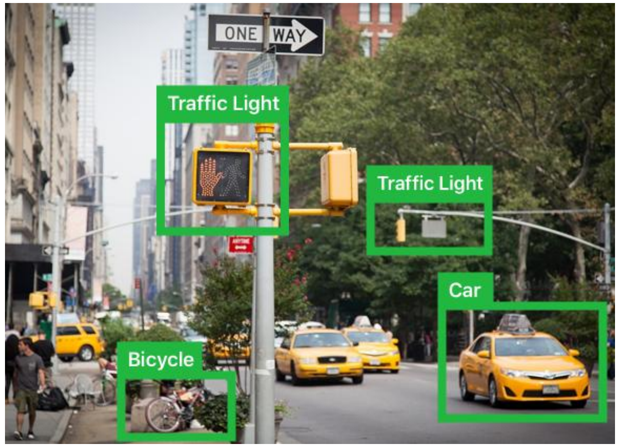
  - 交易欺诈检测



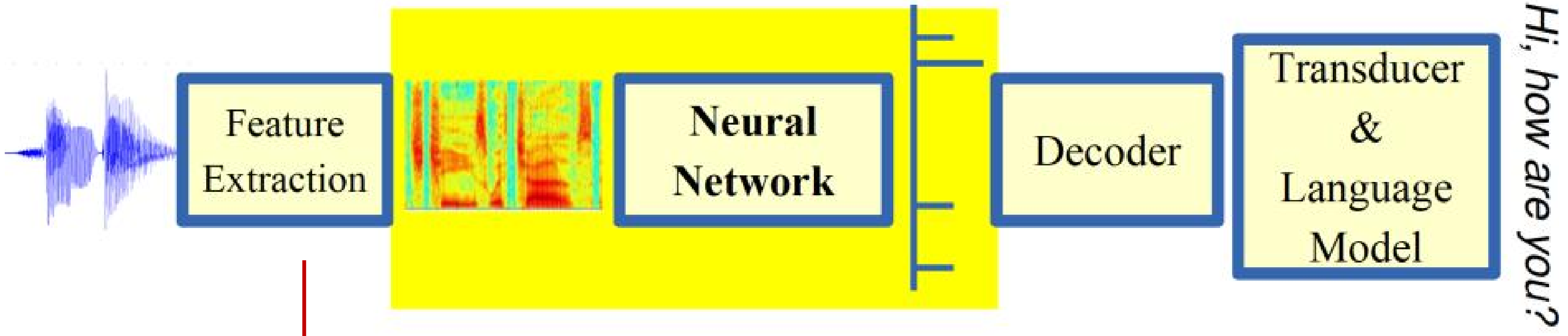
示例

- 目标检测

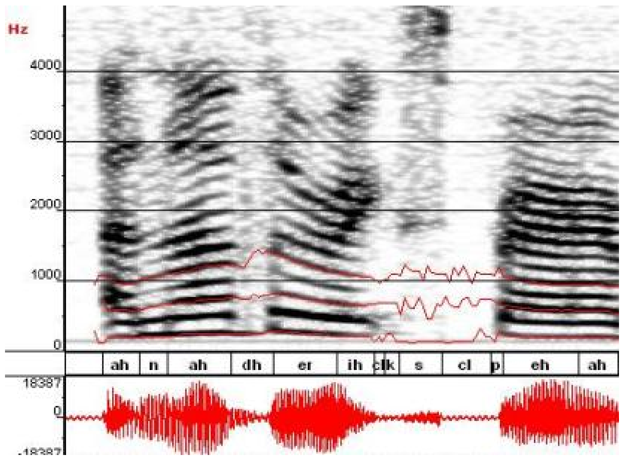

Person
Skateboard


ONE WAY
Traffic Light
Traffic Light
Car
Bicycle

- 语音识别



Feature Extraction
Neural Network
Decoder
Transducer & Language Model
Hi, how are you?

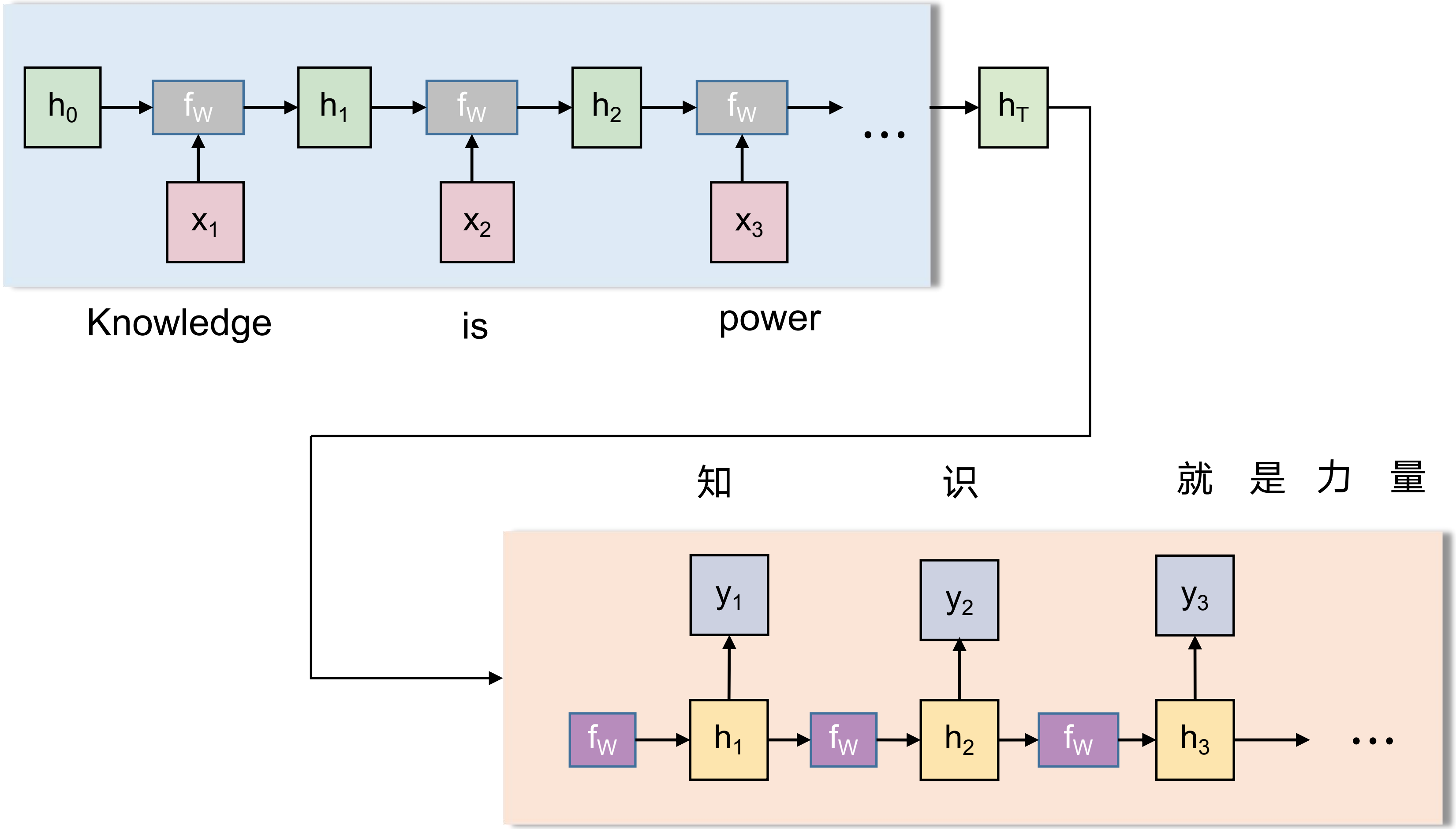


Hz
4000
3000
2000
1000
0
ah
n
ah
dh
er
ih
clk
s
cl
p
eh
ah
18387
0

声谱图

Google now
XBOX! BRING ME A PIE!
DRAGON DICTATION
讯飞语点
您需要做什么?
打开手机中国
正在打开手机中国
点击说话

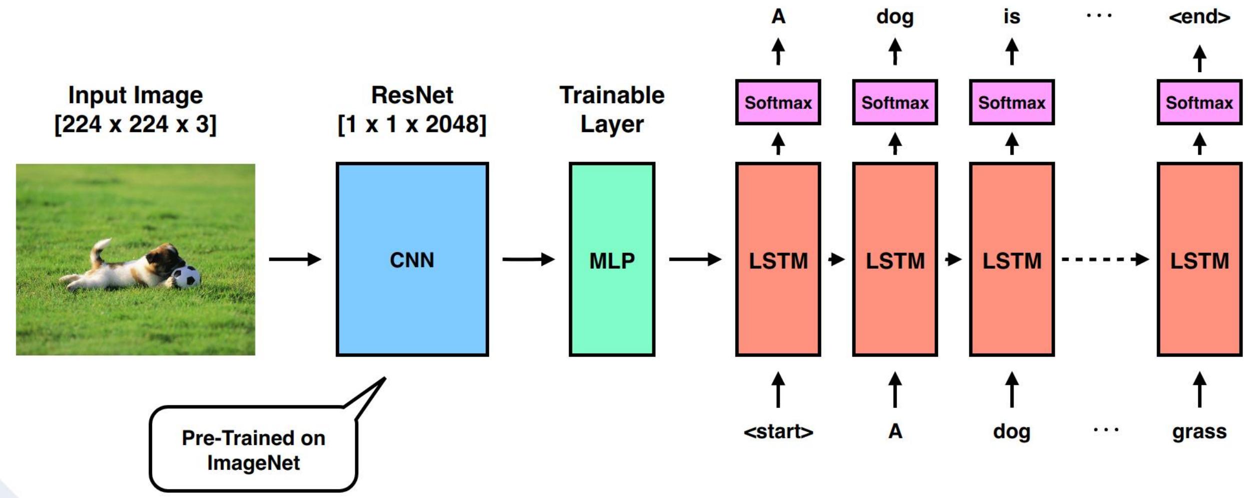
- 神经机器翻译



h0
fW
h1
fW
h2
fW
...
hT
x1
x2
x3
Knowledge
is
power
知
识
就 是 力 量
y1
y2
y3
fW
h1
fW
h2
fW
h3
...

- 图像描述生成

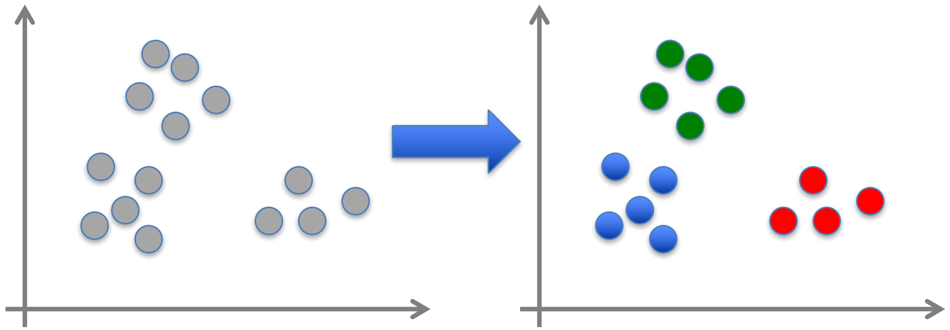

A young boy is playing basketball.
Two dogs play in the grass.
A dog swims in the water.
A group of people walking down a street.
A group of women dressed in formal attire.
Two children play in the water.


Input Image
[224 x 224 x 3]
ResNet
[1 x 1 x 2048]
Trainable Layer
CNN
MLP
LSTM
Softmax
A
dog
is
<end>
<start>
A
dog
grass
Pre-Trained on ImageNet

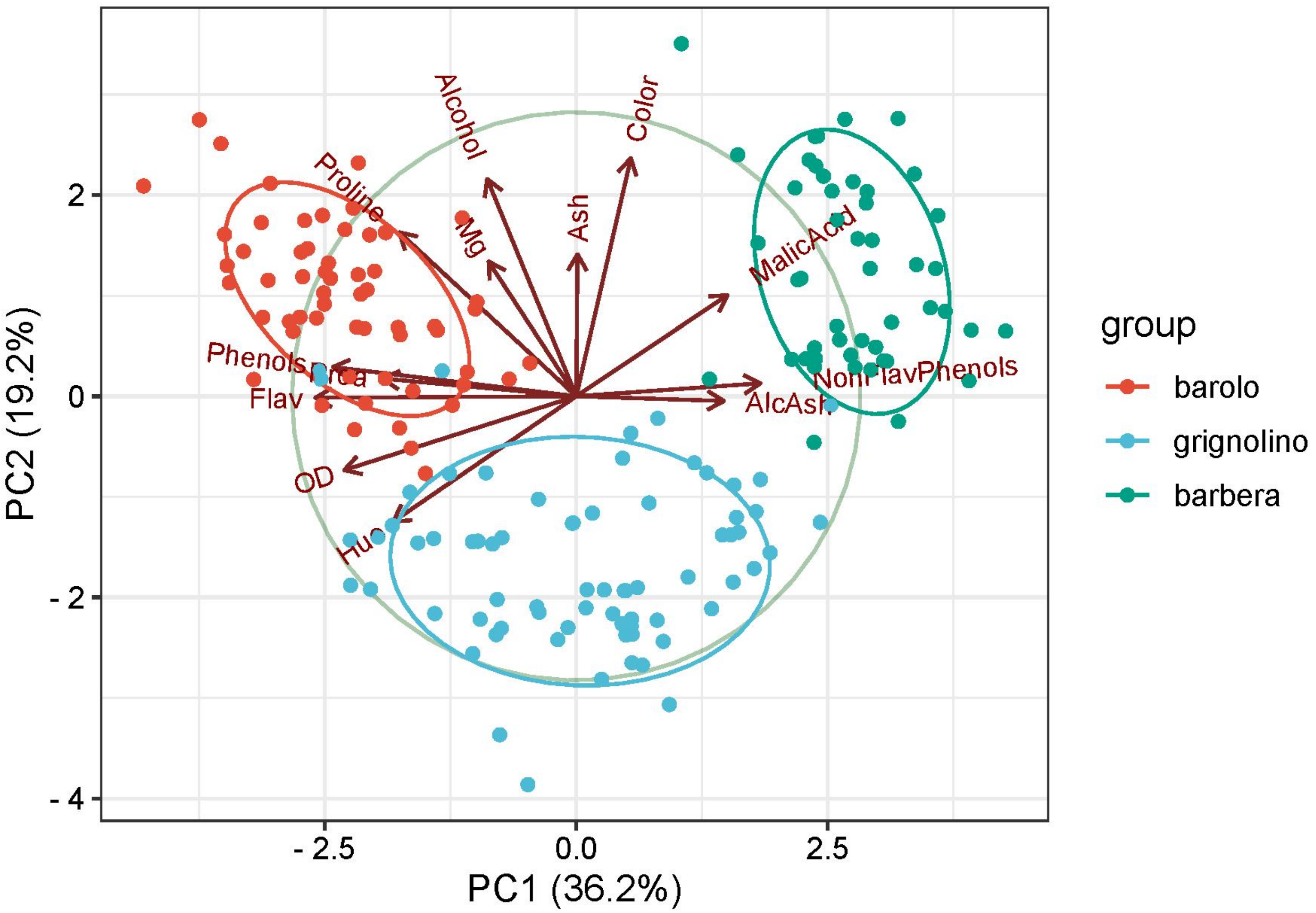
无监督学习

- 给定输入数据 $x_1, x_2, \cdots, x_n$ （不含预测值或标签）输出数据 x_i中隐藏的信息

- 示例：聚类



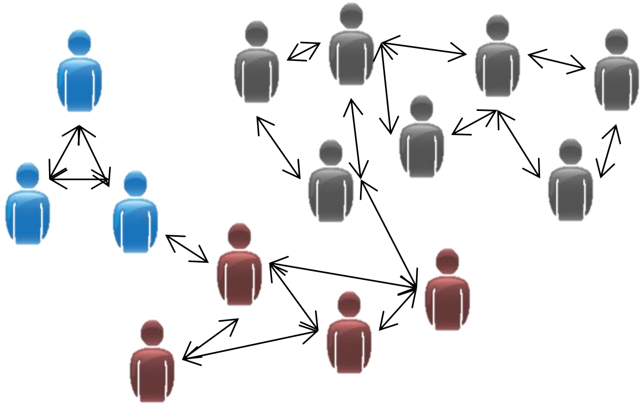
- 降维



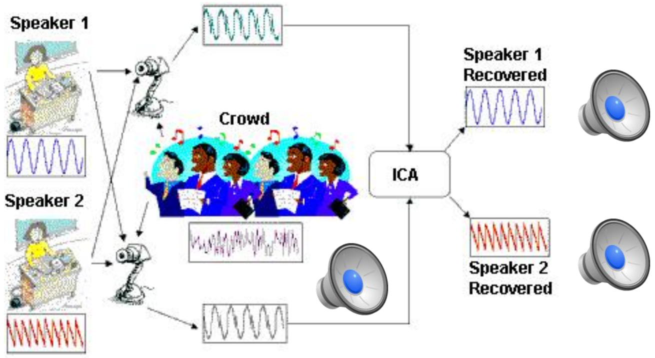
Alcohol
Color
Proline
Mg
Ash
MalicAcid
Phenols
Proa
Flav
NonFlavPhenols
AlcAsh
OD
Hue
2
0
-2
-4
-2.5
0.0
2.5
PC2 (19.2%)
PC1 (36.2%)
group
barolo
grignolino
barbera

- 图像分割

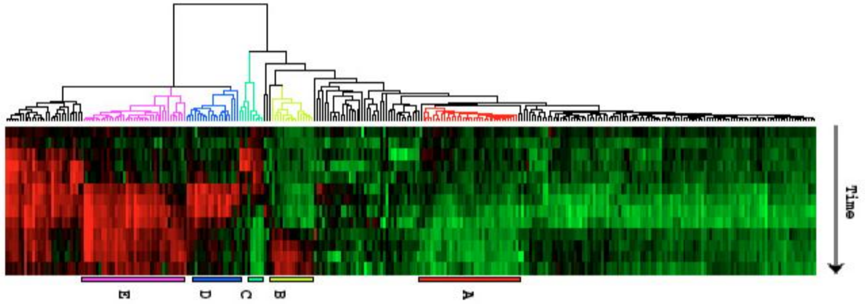
- 社交网络分析



- 分离混合信号

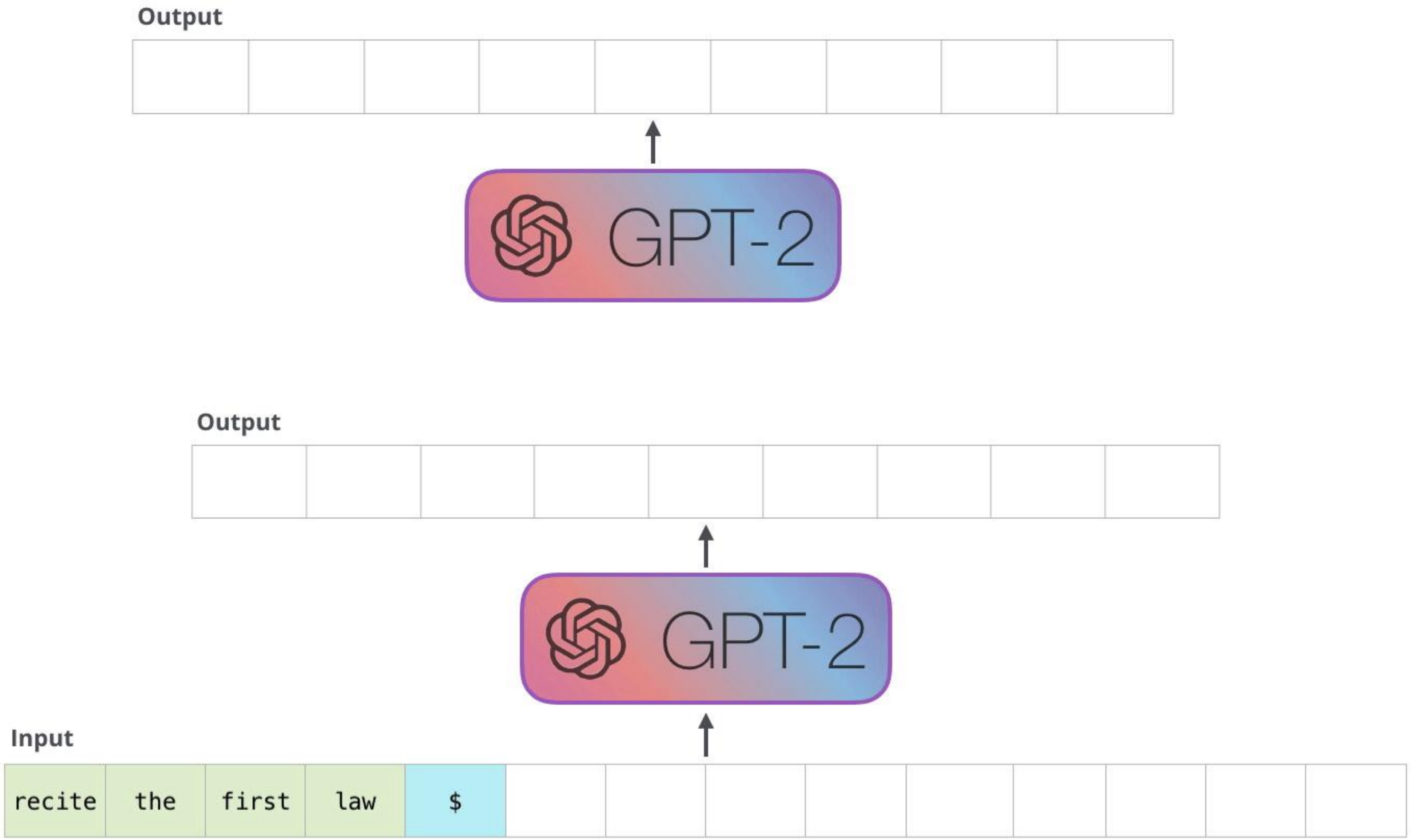


- 基因表达聚类



- 语言模型（自监督）

A robot may not injure a human being



语言模型的应用（可能需要一些额外的弱监督）

- 续写

Python is a programming language that lets you work more quickly and integrate your systems more effectively. You can learn to use Python and see almost immediate gains in productivity and lower maintenance costs. Python runs on Windows, Linux/Unix, Mac OS X, and has been ported to the Java and .NET virtual machines. Python is free to use, even

- 问答

Q Give me five advices on how to learn machine learning

Certainly, here are five key pieces of advice to help you effectively learn machine learning:

1. **Build a Strong Foundation:**
   Begin by understanding the fundamental concepts of machine learning. Study linear algebra, calculus, and probability as they are essential for comprehending algorithms and models. A strong mathematical foundation will make advanced topics easier to grasp.
2. **Online Courses and Tutorials:**
   Enroll in reputable online courses dedicated to machine learning. Platforms like Coursera, edX, and Udacity offer structured and comprehensive courses taught by experts. Andrew Ng's "Machine Learning" course on Coursera is a popular starting point.
3. **Hands-On Practice and Projects:**
   Practical experience is vital. Apply what you've learned by working on projects. Start with basic exercises and gradually progress to more complex problems. Building and experimenting with your models will deepen your understanding.
4. **Explore Diverse Resources:**
   Don't limit yourself to a single resource. Utilize textbooks, online tutorials, blogs, YouTube

Reg

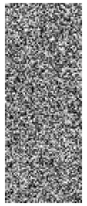
- 聊天机器人
- 改写
- 代码生成

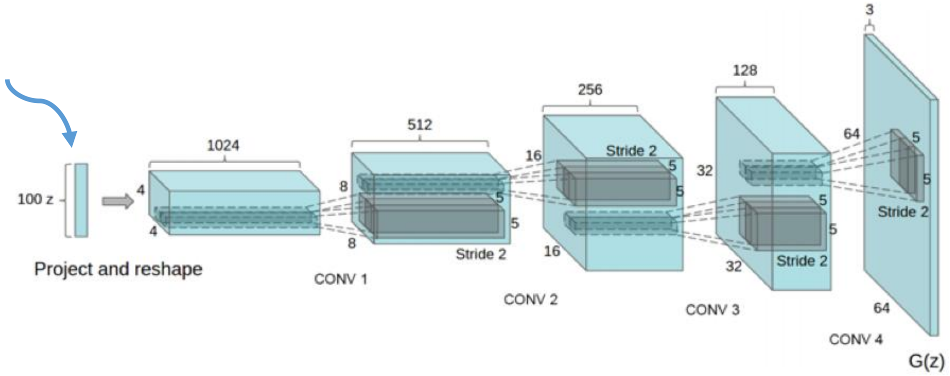
⋮

⋮

- 图像生成模型

噪声



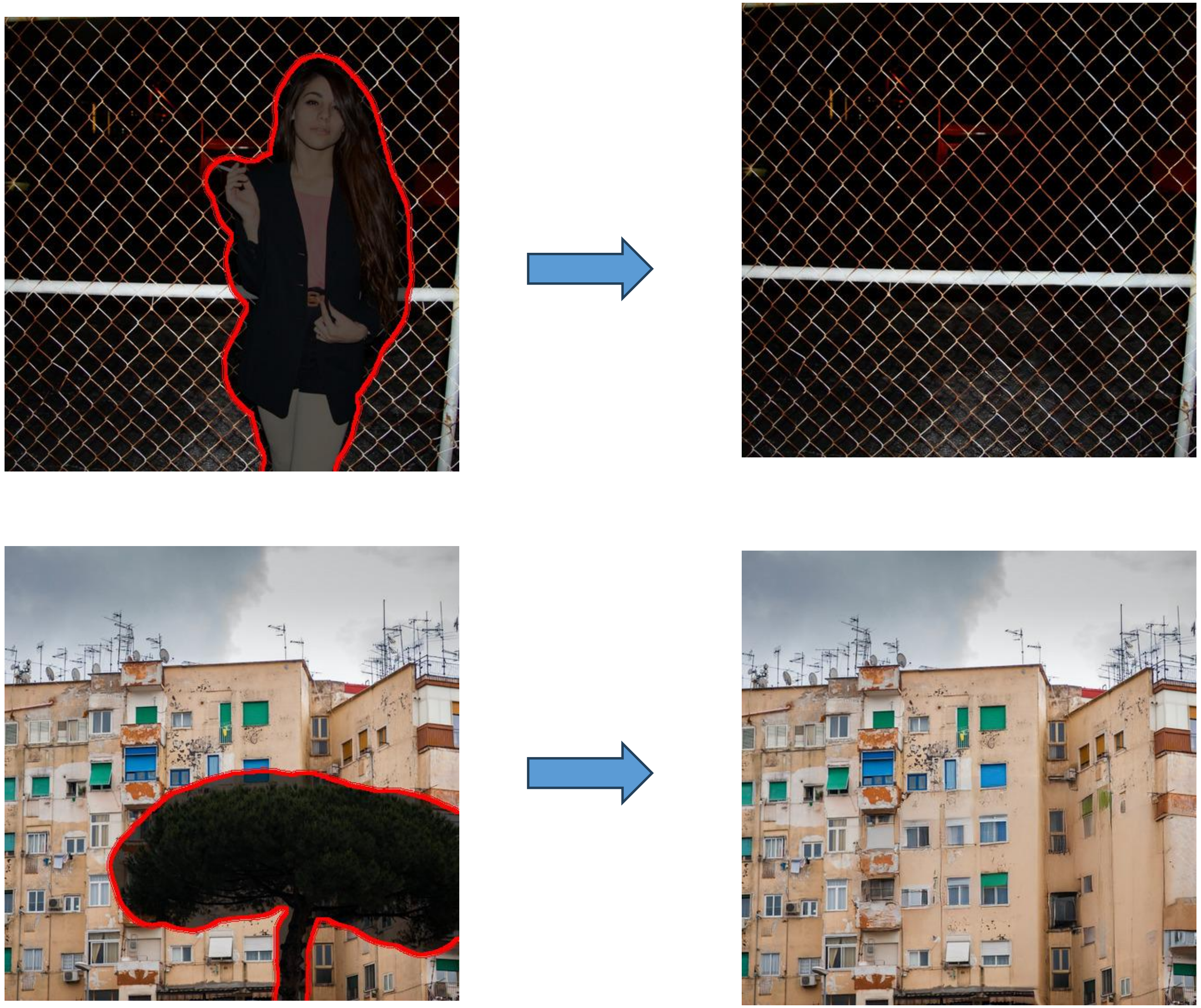


生成器

生成图像

生成模型的应用（可能需要一些额外的弱监督）

– 图像修复



可控生成

文本->
图像

vibrant portrait painting of Salvador Dalí with a robotic half face　　a shiba inu wearing a beret and black turtleneck　　a close up of a handpalm with leaves growing from it

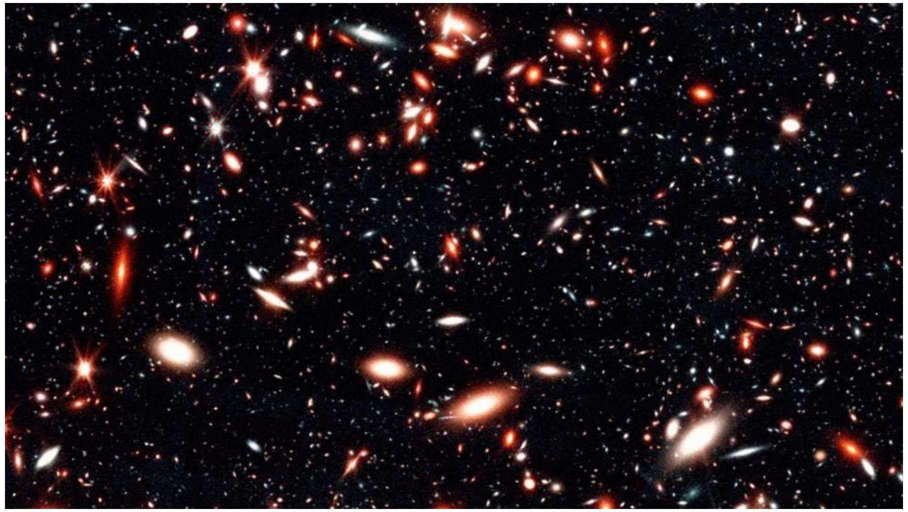
关键点->
图像

"astronaut"

涂鸦 ->
图像

"an elephant with background in the field"　　"Egyptian elephant sculpture"

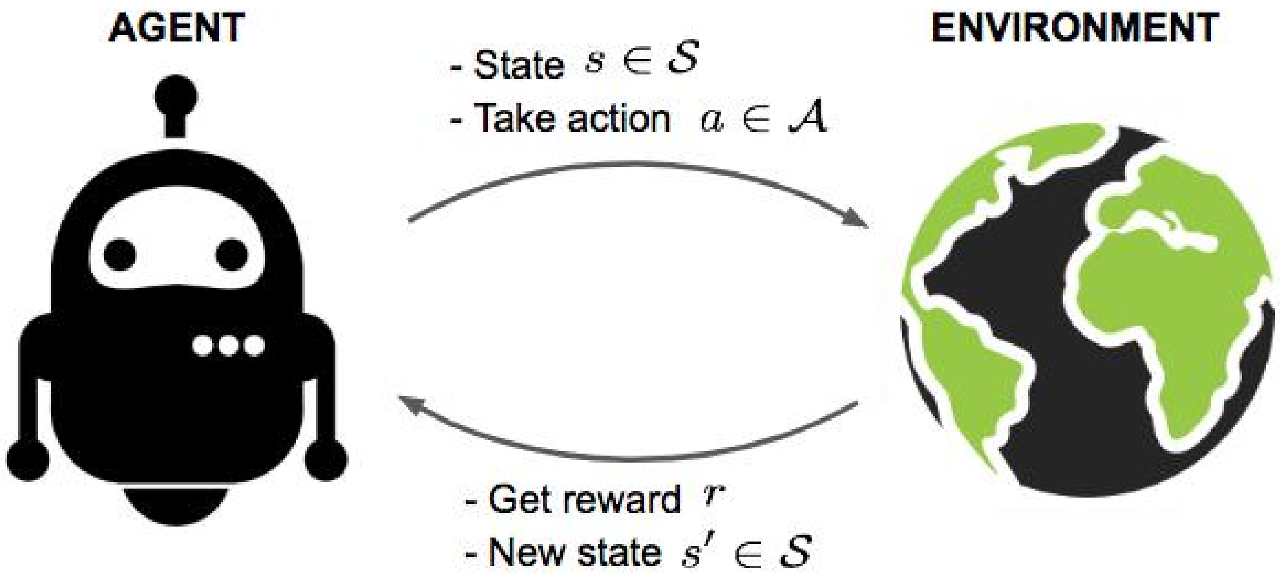
- AI for Science，例如：天文数据分析、药物发现等



James Webb 太空望远镜

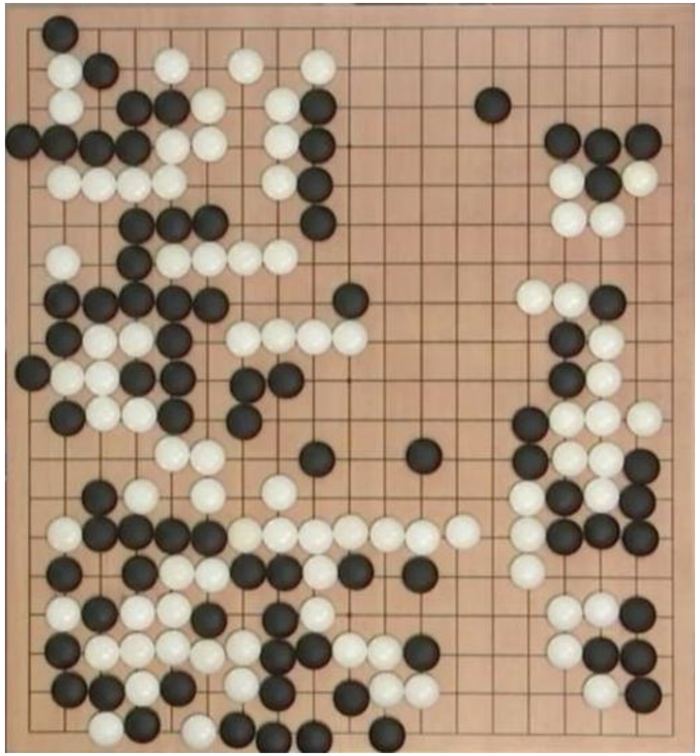
强化学习

- 给定一个状态、动作和奖励序列，学习一个策略，该策略能够指导你*在给定状态下应该采取什么动作*



示例

- 玩游戏



Go

Atari

示例

- 自主直升机

示例

- 基于人类反馈的强化学习

LE

Can you explain to me how RL with human feedback works for language models like I am five?

Sure! So, imagine you have a robot named Rufus who wants to learn how to talk like a human. Rufus has a language model that helps him understand words and sentences.

First, Rufus will say something using his language model. For example, he might say "I am a robot."

Then, a human will listen to what Rufus said and give him feedback on whether it sounded like a natural sentence a human would say. The human might say, "That's not quite right, Rufus. Humans don't usually say 'I am a robot.' They might say 'I'm a robot' or 'I am a machine.'"

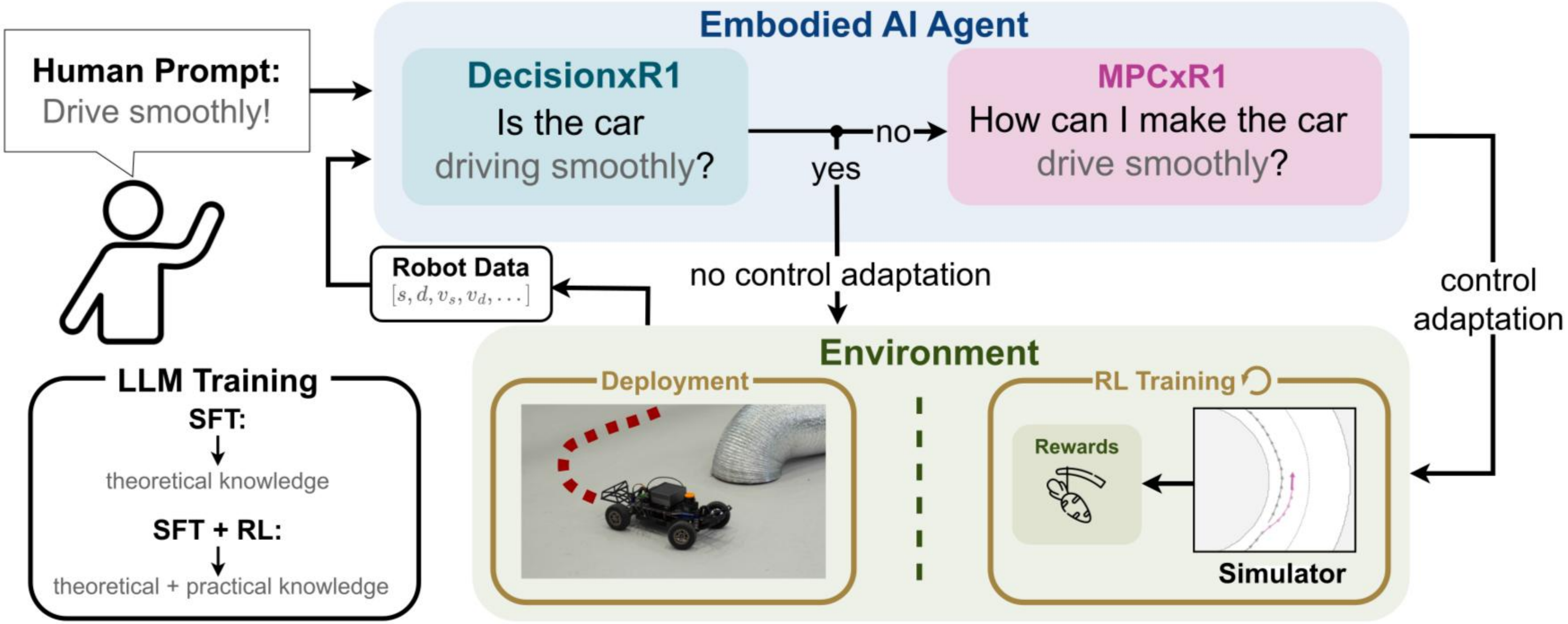
Rufus will take this feedback and use it to update his language model. He will try to say the sentence again, using the new information he received from the human. This time, he might say "I'm a robot."

The human will listen again and give Rufus more feedback. This process will continue until Rufus can say sentences that sound natural to a human.

Over time, Rufus will learn how to talk like a human thanks to the feedback he receives from humans. This is how language models can be improved using RL with human feedback.

示例

- 具身智能



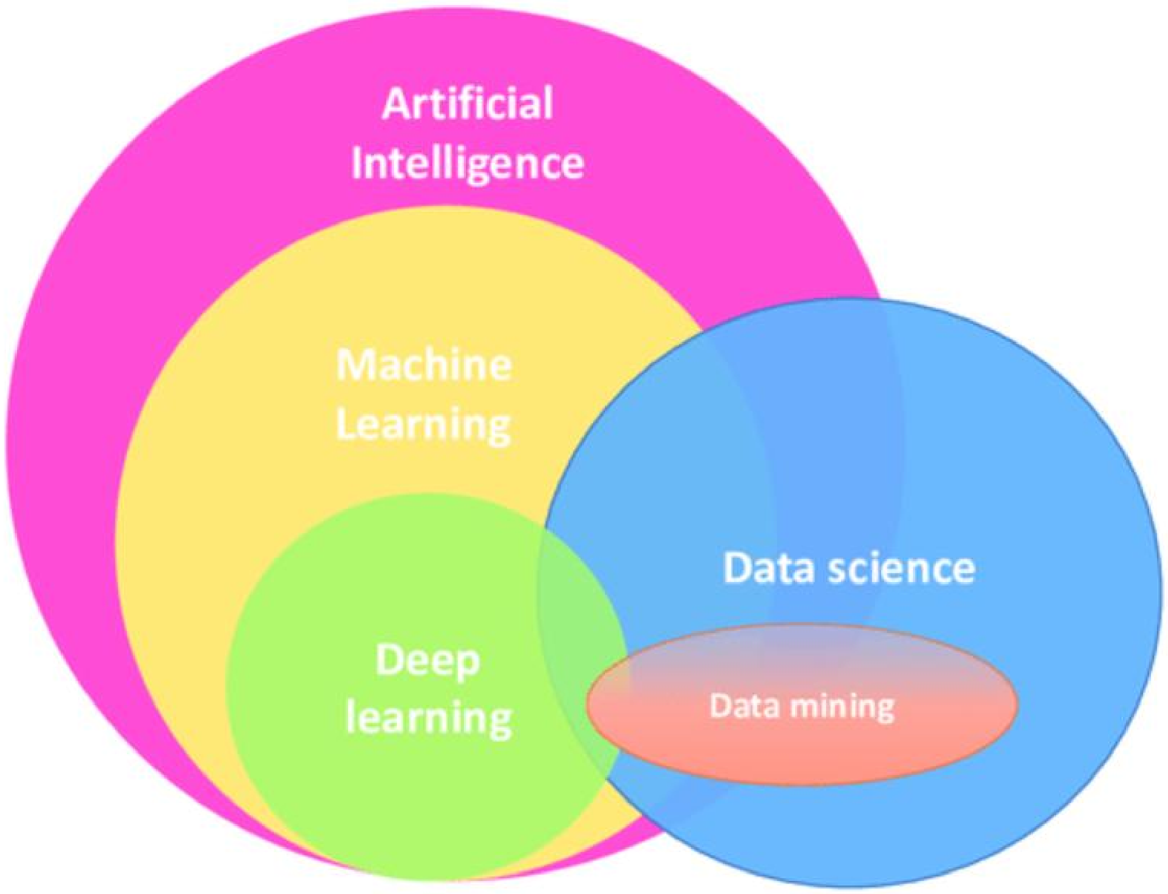
Embodied AI Agent
Human Prompt:
Drive smoothly!
DecisionxR1
Is the car driving smoothly?
no
yes
MPCxR1
How can I make the car drive smoothly?
Robot Data
$[s, d, v_s, v_d, \ldots]$
no control adaptation
control adaptation
Environment
Deployment
RL Training
Rewards
Simulator
LLM Training
SFT:
theoretical knowledge
SFT + RL:
theoretical + practical knowledge

什么是数据挖掘？

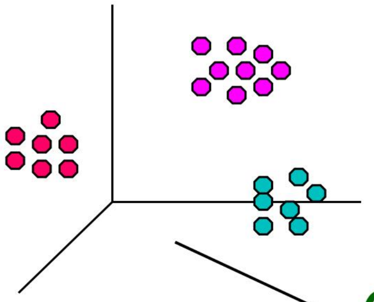
运用机器学习方法从海量数据中发现有用的知识

两种类型 $\begin{cases} \text{描述型} \\ \\ \text{预测型} \end{cases}$

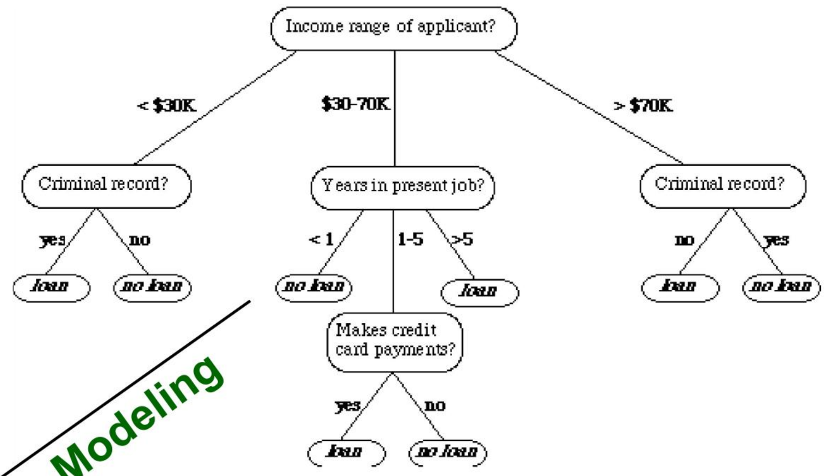
数据挖掘与机器学习的关系



数据挖掘任务



Clustering



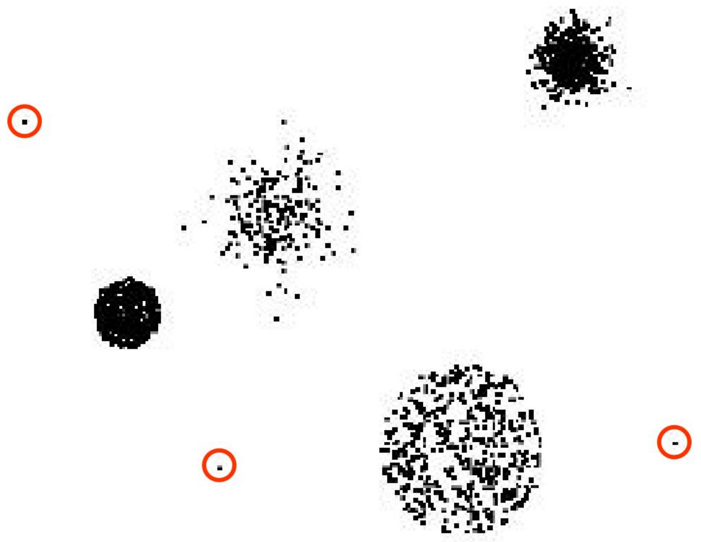
Predictive Modeling

Data

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes
11	No	Married	60K	No
12	Yes	Divorced	220K	No
13	No	Single	85K	Yes
14	No	Married	75K	No
15	No	Single	90K	Yes

Association Rules

Anomaly Detection



两种类型的数据挖掘任务

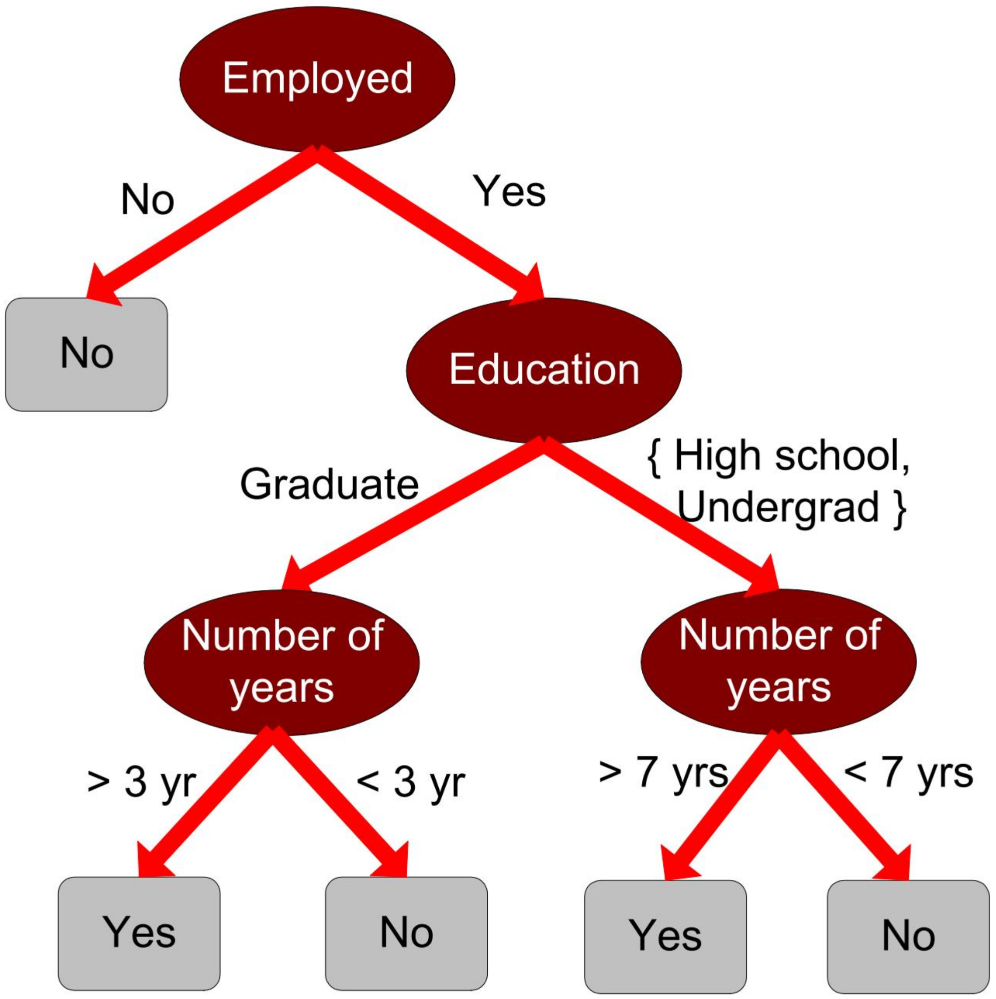
- 预测型：利用部分变量来预测其他变量的未知或未来值
  - 推荐系统
  - 异常检测
  - 类别预测

示例

- 决策树

Class

Tid	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Graduate	5	Yes
2	Yes	High School	2	No
3	No	Undergrad	1	No
4	Yes	High School	10	Yes
…	…	…	…	…



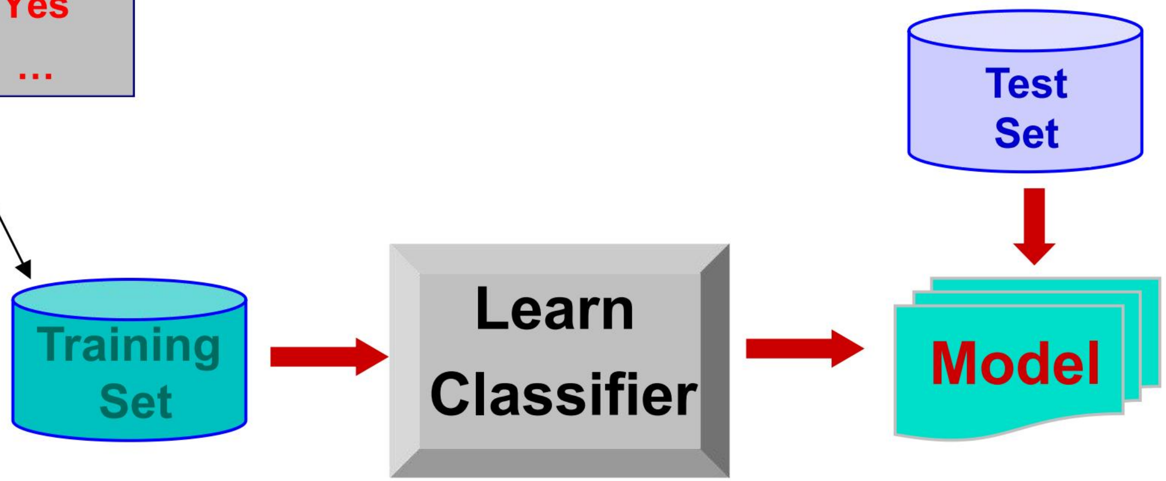
示例

- 分类器

categorical categorical quantitative class

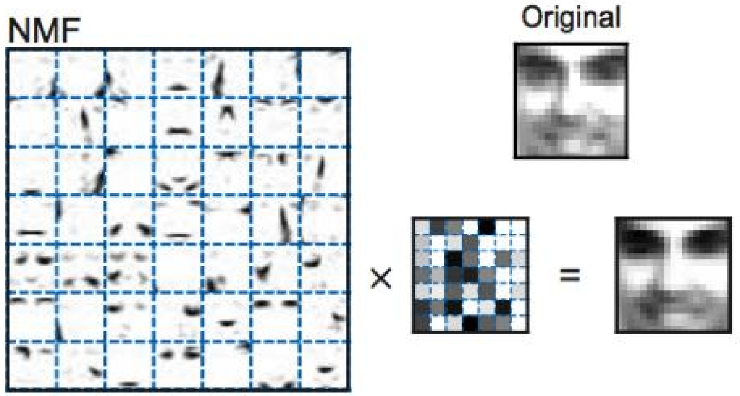
Tid	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Graduate	5	Yes
2	Yes	High School	2	No
3	No	Undergrad	1	No
4	Yes	High School	10	Yes
…	…	…	…	…

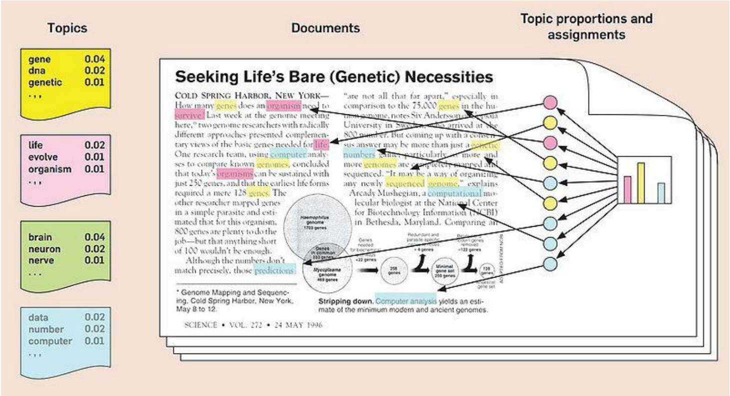
Tid	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Undergrad	7	?
2	No	Graduate	3	?
3	Yes	High School	2	?
…	…	…	…	…



两种类型的数据挖掘任务

- 描述形：从数据中发现人类可理解的隐藏模式
  - 聚类
  - 可解释的表征（例如：非负矩阵分解、主题模型）





关联规则

- 依赖关系规则

TID	*Items*
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

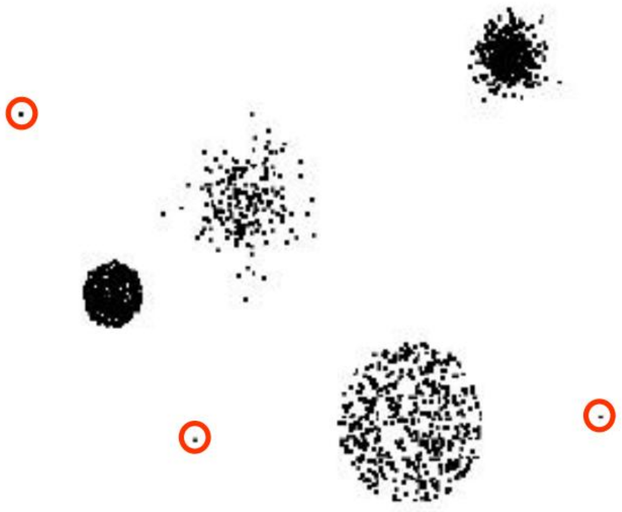
Rules Discovered:

{Milk} --> {Coke}

{Diaper, Milk} --> {Beer}

异常检测

- 检测与正常数据的显著偏差



本课程内容概览

- **机器学习基础**
  - ✓ 线性回归
  - ✓ 线性分类
  - ✓ 支持向量机
  - ✓ 基于树的方法
  - ✓ 聚类：K-均值
  - ✓ 神经网络
- **感知与推理**
  - ✓ 线性降维：主成分分析（PCA）
  - ✓ 图像表征学习
  - ✓ 文本表征学习
  - ✓ 多模态表征学习
- **机器生成**
  - ✓ 潜变量模型
  - ✓ EM算法
  - ✓ 变分自编码器
  - ✓ 扩散模型
  - ✓ 可控生成与AIGC
  - ✓ ………
- **数据挖掘实践**
  - ✓ 异常检测
  - ✓ 推荐系统
  - ✓ ……